

View results

Respondent

64

Anonymous

75:51

Time to complete

A. General information about the respondent

1. * Country

Mexico



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Secretariat of Science, Humanities, Technology and Innovation

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://secihti.mx/>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The Secretariat of Science, Humanities, Technology, and Innovation (Secihti) is the institution of the Government of Mexico (2024-2030) that formulates and guides national policy on the subject. It articulates and coordinates the capacities, knowledge, resources, and talent of researchers and technologists to consolidate a National System of Science, Humanities, Technology, and Innovation.

As the head of the sector, Secihti oversees the Mexican State in the Humanities, Science, Technology, and Innovation, and contributes to educational policy, primarily that of higher education, in coordination with the Ministry of Public Education (SEP).

The Secretariat designs and executes strategic projects through two Undersecretariats: Science and Humanities and Technological Development, Outreach, and Innovation, in addition to a Cross-Cutting Policies Unit. These bodies contribute to collaboration, understanding, and the addressing of national priorities through an inter-secretarial, inter-sectoral, and inter-institutional working model.

7. * Full name of officially designated contact person(s) who completed this survey

Zaira Lagunas Ledesma

8. * Position of officially designated contact person(s) who completed this survey

Subdirector

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Zaira Lagunas Ledesma

11. Brief description of the consultation process established for the preparation of the report

A thorough analysis was conducted of portals, microsites, repositories, and other digital sources that provided information relevant to the questions included in the report requested by UNESCO. In addition, in some cases, support was requested from other departments within SECIHTI to supplement the required data.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

None

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

None

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

None

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- ☐ Public
- ☐ Private
- ☐ Civil Society
- ☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN
SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO
OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- ☒ Yes
- ☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Yes, UNESCO's recommendation on Open Science was disseminated through a series of six webinars on the official YouTube channel, in which different actors from the humanities, science, technology and innovation community shared their experiences and knowledge on: 1. Open Scientific Knowledge https://www.youtube.com/watch?v=Xbg-pOv_YZg&list=PLUXfjz1LI_DKiRi9oBKTdTFQpl3drkJBq 2. Scientific Publications and Open Educational Resources https://www.youtube.com/watch?v=QfyHNzTfG64&list=PLUXfjz1LI_DKiRi9oBKTdTFQpl3drkJBq&index=2 3. Research Management Systems (CRIS) https://www.youtube.com/watch?v=IrCgRiR-n5I&list=PLUXfjz1LI_DKiRi9oBKTdTFQpl3drkJBq&index=3 4. Citizen Science and Gender Equality https://www.youtube.com/watch?v=BG32AmPZmQo&list=PLUXfjz1LI_DKiRi9oBKTdTFQpl3drkJBq&index=4 5. Copyright in the framework of Open Science and Plagiarism and Academic Citation https://www.youtube.com/watch?v=hvBw7mLNdnM&list=PLUXfjz1LI_DKiRi9oBKTdTFQpl3drkJBq&index=5 6. Open Science today https://www.youtube.com/watch?v=GixDB05KUy0&list=PLUXfjz1LI_DKiRi9oBKTdTFQpl3drkJBq&index=6

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Spanish

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Yes, there is an expectation that some elements of the key components of the Open Science Recommendation can be promoted through national systems and other actions to be undertaken by the Ministry of Science, Humanities, Technology and Innovation.

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Yes, it will be encouraged that the results of research that has been funded in whole or in part by the public sector can be registered in open access computer repositories.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

25. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Yes, it is called the General Law on Humanities, Sciences, Technologies, and Innovation, and its aim is to guarantee the exercise of the human right to science in accordance with the principles of universality, interdependence, indivisibility, and progressivity. This is so that everyone may enjoy the benefits of scientific development and technological innovation, as well as human rights in general. https://www.dof.gob.mx/nota_detalle.php?codigo=5688048&fecha=08/05/2023#gsc.tab=0

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☐ open science infrastructure
- ☐ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☐ Yes
- ☒ Partly
- ☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

Yes, with the publication of the General Law on Humanities, Sciences, Technologies, and Innovation, it was decided to employ the term "open access". This term encompasses the guiding values and principles of the UNESCO recommendation, for example, in Article 9: "To fulfil the purpose of this Law, the Federation, federated entities, municipalities, and demarcations, through their corresponding institutions, must implement public policies aimed at carrying out, promoting, and supporting training, research, dissemination, outreach, and the development of projects in the fields of humanities, sciences, technologies, and innovation, as well as open access to the information derived from these activities. The aim is to contribute to the advancement of universal knowledge, the strengthening of national sovereignty, the comprehensive and sustainable development of the country, the well-being of present and future generations, the preservation, restoration, protection, and improvement of the environment, and the achievement of the constitutional objectives of the Mexican State."

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☒ Yes
- ☐ Under development
- ☐ No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Yes, in the Specific Guidelines for the Integration and Operation of Repositories, the importance of registering information products in repositories is addressed, as well as the recognition of FAIR principles; and the recognition of datasets to reinforce the development of research, aspects that in a way can be placed in the pillar of 'open scientific knowledge'.
https://dof.gob.mx/nota_detalle.php?codigo=5738516&fecha=06/09/2024#gsc.tab=0

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☒ Yes
- ☐ Under development
- ☐ No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

As a strategy, Conahcyt, now Secihti, promoted the use of repositories for the registration of open access research products, so that members of the Humanities, Sciences, Technologies, and Innovation community could use or reuse certain content to generate new knowledge, always with full respect for property rights and using open licenses as a protection mechanism for authors. https://dof.gob.mx/nota_detalle.php?codigo=5738516&fecha=06/09/2024#gsc.tab=0

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)



Yes



Under development



No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

Yes, for example, in the case of the National Autonomous University of Mexico, there are General Guidelines for the UNAM Open Access Policy, the objective of which is to promote open access and free consultation of digital content via Internet, resulting from the academic, scientific, research, and cultural activities carried out at UNAM.

<https://www.unamenlinea.unam.mx/recurso/82924-lineamientos-generales-para-la-politica-de-acceso-abierto-de-la-unam>

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☐ Yes
- ☐ Under development
- ☒ No

36. 2.6 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

No, to date, no specific funding strategies have been developed for Open Science. However, with support from the National System of Researchers, scholarships, and other calls for proposals, beneficiaries have been asked to register the products derived from their research projects in open-access repositories.

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

Yes, under the General Law on Humanities, Sciences, Technologies and Innovation, Article 3, Section IV, states that part of the objectives of the National Council Law, now SECIHTI, includes establishing "public mechanisms and instruments to provide sufficient resources and incentives to promote and support training, research, dissemination, and project development in the humanities, sciences, technologies, and innovation, as well as open access to information derived from such activities." In this way, through the Calls for Proposals, it has been promoted that information derived from research is guaranteed to be openly accessible.

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

40. 2.8 cont. **If yes**, please provide details and links as relevant.

Yes, it is considered that by signing an agreement with LA Referencia, the recognition of humanistic, scientific, technological, and innovative production that has received public funding in Mexico for its respective development can be promoted. Through this collaboration, the research outputs that are being generated daily by the national repository will be disseminated.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☒ Yes
- ☐ Under development
- ☐ No

42. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

Yes, in the period 2021-2024, in the document "Progress and Results" of the Special Programme on Humanities, Sciences, Technologies, and Innovation 2021-2024, under "Priority Strategy 6.4 Establishing open access policies to science for scientists, technologists, and citizens regarding research, development, and innovation results carried out with public resources, whether they be data, reports, or documents, thus generating greater reintegration of knowledge into research" (p. 101-102; 141-144), the progress towards the goal is shared, and in a way, through these reports, the incidence and monitoring related to open access is documented

https://www.gob.mx/cms/uploads/attachment/file/946761/38_PEHCITI_AyR2324.pdf

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

43. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

According to the National Institute of Statistics and Geography (INEGI) in Bulletin Indicator 90/25, information is shared regarding the variation of the Quarterly Gross Domestic Product (GDP) as of 21 February 2025; this document refers to "professional, scientific, and technical services."
https://www.inegi.org.mx/contenidos/saladeprensa/boletines/2025/pibt/pib_Pconst2025_02.pdf

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

According to the National Institute of Statistics and Geography (INEGI) in Press Release No. 299/24 dated May 15, 2024, it is mentioned that in 2022, according to information from ENDUTIH, "68.5% of households in Mexico had internet access (25.8 million households) [...]."
https://inegi.org.mx/contenidos/saladeprensa/aproposito/2024/EAP_DMInternet.pdf

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

46. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

Yes, through the National Red Ecosystem, which is "a collaborative model that incorporates joint projects with an interinstitutional, transdisciplinary and multidisciplinary approach, thus contributing, through knowledge, to the understanding, attention and resolution of local, communal, regional, national and international priorities". Such networks will be a priority for SECIHTI so that working groups can be formed focusing on topics such as: health sciences, science and society, sustainable development and climate change, education, risk management, communication, and internationalisation, among others." <https://secihti.mx/red-ecos-nacional/>

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☒ Yes
- ☐ Under development
- ☐ No

48. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
- ☒ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

49. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

The National Council, now SECIHTI, developed the National Repository is identified as part of the platforms for knowledge exchange and co-creation. This tool concentrates open-access, full-text scientific production from the various institutional repositories that have received public funding for the development and implementation of their platforms. It is worth mentioning that the new version of this network will be launched soon.

<https://repositorionacionalcti.mx/> <https://repositorionacionalcti.mx/documentos>
https://dof.gob.mx/nota_detalle.php?codigo=5738516&fecha=06/09/2024#gsc.tab=0 At the present, the SECIHTI, through the ECOS Network, and initiative of the Secretariat of Education, Science, Technology and Innovation (Sectei) of Mexico city, aims to create a space for interinstitutional, multidisciplinary and multisectoral collaboration, from which proposals, solutions and strategies can emerge for the benefit of the population, through thematic, territorial and interinstitutional strategic alliances. In this sense, and as part of the examples of its implementation, the Network "Artificial Intelligence and High Performance Computing" is in the process of consolidation, of which a meeting with different institutions was generated and of which the following evidence is shared: https://www.youtube.com/live/wLfmRqumXrY?si=1LtOK-qFNSx9_Y_R

50. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☒ Yes

☐ No

51. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Yes, through National Laboratories, which, according to the "UNESCO Recommendation on Open Science," can be categorized as "open science infrastructures." As such, they are considered inter-institutional infrastructures distributed throughout the country and seek to influence the generation of scientific, technological, and innovation capacities. This serves as a mechanism to strengthen HCTI communities. <https://secihti.mx/conahcyt/areas-del-conahcyt/desarrollo-cientifico/laboratorios-nacionales/>

52. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒ Yes

☐ No

53. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

In this regard, the "Conacyt National Laboratories (LNC)" have been created, inter-institutional infrastructures distributed throughout the national territory that seek to influence the generation of scientific, technological, and innovation capacities. Since they are public entities, they are non-profit and serve social welfare." To consult the Registry of Beneficiaries of the National Laboratories Program, see <https://secihti.mx/convocatorias/convocatorias-programa-de-laboratorios-nacionales/padron-de-beneficiarios-programa-de-laboratorios-nacionales/> <https://secihti.mx/conahcyt/areas-del-conahcyt/desarrollo-cientifico/laboratorios-nacionales/>

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

54. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☐ Yes

☒ No

55. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

To date, the Secretariat has not implemented concrete actions—such as surveys or other data-gathering tools—aimed at determining open science capacities in the country's higher education institutions. However, the possibility of including this initiative in the institutional agenda is being considered.

56. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

57. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, to date, no capacity-building program for the integration of fundamental values and principles of open science has been implemented. However, SECIHTI, in each and every call for proposals, refers to the basic principle that everyone has the right to enjoy the benefits of the development of science and technological innovation and establishes that the State is obligated to support scientific, humanistic, and technological research and innovation, as well as guarantee open access to the information derived from it. Likewise, these types of documents promote inclusion by prioritizing proposals from researchers belonging to underrepresented communities and vulnerable groups, including Indigenous and Afro-descendant peoples, people living with disabilities, women, and people of color. These aspects, together, allude to values such as "diversity and inclusion," "equity and justice," and "collective benefits" within the framework of the principles and values of open science. For an example of this type of call, see: https://secihti.mx/wp-content/uploads/convocatorias_secihti/convocatorias-ciencia-y-humanidades/ciencia-basica-frontera/2025/Conv_CDF2025_VF.pdf

58. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

59. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, so far no actions have been implemented to include open science competences in the curricula, mainly because each educational institution has a certain autonomy in this area. However, the recognition of tools such as the repository and other applications developed by SECIHTI that take into account the principles and values of open science can be promoted.

60. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

☐ Yes

☒ Partly

☐ No

61. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

☒ Yes

☐ No

62. 4.5 cont. **If yes**, please provide more information and links as relevant.

area and indicate if there are plans to do so. Yes, the Conahcyt, now the SECIHTI, once promoted a "Conference on Scientific Communication and Universal Access to Knowledge". Its aim was to bring knowledge in the humanities, sciences, technologies and innovation closer to society as a mechanism to guarantee the human right to science for the benefit of Mexico. <https://secihti.mx/evento/jornadas-de-comunicacion-cientifica-y-acceso-universal-al-conocimiento/>

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

63. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☐ Yes

☒ No

64. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, however, we recognize the efforts of entities such as UNAM, which have promoted Open Access policies (<https://www.unamenlinea.unam.mx/recurso/82924-lineamientos-generales-para-la-politica-de-acceso-abierto-de-la-unam>) that are aligned with the principles and values of Open Science. We also recognize some actions that the Autonomous Metropolitan University (<https://boletines.uam.mx/archivos/463-23/>) has designed as a mechanism for adopting the relevant culture. Additionally, these topics will be analyzed for possible inclusion in the agenda.

65. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☒ Yes

☐ No

66. 5.2 cont. **If yes**, please provide details and links as relevant.

Yes, as part of the bases for formulating, executing and evaluating public policies, Article 11, Section XVIII establishes the "construction and development of a humanistic, scientific, technological and innovation culture based on epistemological rigor, dialogue of knowledge, horizontal and transversal production of knowledge, epistemic plurality and equity, interculturality, collaborative work and the vindication of the humanities [...]." As part of the application examples, the "Analysis Forum. The Mayan milpa, productive and social ecosystem: dialogue of knowledge and impact in southeastern Mexico" is shared, which in addition to sharing experiences on the subject, promoted the articulation of projects in favor of the communities of the region, it is very likely that these types of elements will be considered in the present and future actions of the agenda, which are aligned with the practices of open science. For further information, see: <https://secihti.mx/evento/foro-de-analisis-la-milpa-maya-ecosistema-productivo-y-social-dialogo-de-saberes-e-incidencia-en-el-sureste-mexicano/> https://www.dof.gob.mx/nota_detalle.php?codigo=5688048&fecha=08/05/2023#gsc.tab=0

67. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes

☒ No

68. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No. To date, no specific initiatives have been established in this area, nor are their implementation planned for the end of 2025. However, the Secretariat remains committed to evaluating the feasibility of such measures within its science policy agenda.

69. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

70. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (*ref.: (vi) 21*)

- ☐ YES, at the national level
- ☒ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☐ No

71. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☐ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☐ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☐ promoting open innovation practices
- ☐ Other

72. 6.1. cont. **If yes**, please provide more information and links as relevant

Yes, SECIHTI, with the possibility of having a complete map of the different stages of the research cycle, promoted the use of technical sheets in ENI SALUD projects (<https://repositorio-salud.conacyt.mx/jspui/handle/1000/38>) as a strategy to record the data collection process, an element that was not previously implemented in other projects. However, this practice is recorded in the Specific Guidelines for the Integration and Operation of Repositories, in Article 21, which states: "The metadata for accessing, searching, reading, and downloading full-text information and datasets in digital formats applicable to computer repositories will be established in the Repository Manual."
https://dof.gob.mx/nota_detalle.php?codigo=5738516&fecha=06/09/2024#gsc.tab=0

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

73. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

The General Law on Humanities, Science, Technology and Innovation makes open access to publicly funded research mandatory. This legal requirement (set out in Article 56) ensures that the results of publicly funded research are openly available, which in itself facilitates international access and potential collaboration. In addition, the Specific Guidelines for the Integration and Operation of Repositories place particular emphasis on ensuring that records of information products in repositories comply with the FAIR (Findable, Accessible, Interoperable and Reusable) principles to facilitate communication with other platforms and speed up information retrieval by the global scientific community, in line with the objectives of the UNESCO Recommendation. <https://www.diputados.gob.mx/LeyesBiblio/pdf/LGMHCTI.pdf>

74. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☒ Yes

☐ No

75. 7.2 cont. **If yes**, please provide links as relevant and more information.

Yes, partially. Article 10, Section VI of the Internal Regulations of the Secretariat of Science, Humanities, Technology, and Innovation establishes that the Undersecretariat of Science and Humanities is responsible for promoting inter-institutional collaboration, as well as inter- and multidisciplinary research groups, with various sectors of society, to stimulate basic and cutting-edge research in all fields of knowledge and strengthen national humanistic research agendas. This aspect supports the General Law on Humanities, Science, Technology, and Innovation in guaranteeing the human right to science and open access to information derived from state-funded research. https://www.dof.gob.mx/nota_detalle.php?codigo=5747752&fecha=24/01/2025#gsc.tab=0 https://www.dof.gob.mx/nota_detalle.php?codigo=5688048&fecha=08/05/2023#gsc.tab=0

76. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

77. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, not at this time, but considering the significance of these elements, the feasibility of including this item in the Secretariat's agenda will be analyzed.

Part 8. GENERAL CONSIDERATIONS

78. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

79. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org